Predicting Desirable States

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Synopsis

In an attempt to provide people with more information to make better educated decisions regarding relocating and investing in different states across the United States, this research study provides a statistical model that is intended to be used to predict whether or not a state is likely considered to be more desirable.

Introduction

Question & Background

What is the best place in the U.S. to settle down? To start your career? To retire? Each of these questions can be answered similarly, by identifying which state has the best quality of living. Quality of living can be broken down into a multitude of sub-categories, each of which plays an important role in what differentiates each state from one another.

In 2022, 28.2 million Americans moved, most of which staying local to where they already lived, with 19.9% of those people relocating to a new state entirely (Pelchen). In 2025 Q1, investors were known to be responsible for nearly 27% of all homes sold, the highest percentage share of all quarters over the past 5 years (Veiga). This percentage would equate to roughly 265,000 homes across the country being bought by investors. In order to maximize their future return on invest, and ability to fill their investment properties with tenants, it is important they identify the areas people are looking to reside.

As someone who has moved multiple times in their life, across state lines and within them, and has debated the idea of doing so again in the near future, providing myself and others with as much information as possible in order to do so would be a great assistance in helping make a much more informed decision.

The intent of this study is to identify the most desirable States in the U.S. for residents, based off of a combination of different cost, infrastructure, and safety statistics and more. This information is intended to assist people in making more informed decisions when they are faced with an opportunity and/or decision on whether or not to move.

Hypothesis & Prediction

Hypothesis: Utilizing the quality of life variables pertaining to physical and mental heath we will be able to predict which states would be considered top ten overall states.

Prediction: The quality of life fields, such as mental health counseling and dental visits, will be the only statistically significant in being able to predict whether or not a state is considered to be within the top 10.

Data

Data Acquisition

The data included within the final table is sourced from 9 unique sources, some of which include multiple sources within their tables. The sources for each of tables are Bureau of Economic Analysis in the U.S. Department of Commerce, the National Center for Education Statistics in the Institute of Education Services, America's Health Rankings in the United Health Foundation, the Bureau of Transportation Statistics in the United States Department of Transportation, the FBI Crime Data Explorer from the FBI Uniform Crime Reporting Program, the Missouri Economic Research and Information Center, the U.S. Bureau of Labor Statistics, the United States Census Bureau, and the U.S. News and World Report. Some of these tables include datapoints retrieved from multiple additional sources, which is specified within the data dictionary that can found in the Appendix. All of the data fields utilized in this study are either directly sourced from government sites or are a combination of multiple data points that are each sourced from government sites.

The aforementioned sources and their datapoints combine to form a great final dataset for the intended research as they cover multiple different aspects of what people think of while judging states and their populations. The dependent variable in our model testing is Overall State Rank, sourced from the U.S. News and World Report, was originally formulated using multiple aspects with their underlying data consisting of the categories Education, Health Care, Economy, Infrastructure, Opportunity, Fiscal Stability, Natural Environment, and Crime & Corrections. Being aware that the response variable we would be working with was constructed in this manner, it was integral to the approach of our model to ensure that we attempted to represent the same multi-perspective approach to the overall success of a state.

Cleaning, Pre-processing, and Dimensionality Reduction

The first most important step in setting up our data for eventual model testing was to ensure completeness of the dataset and each of its parts, as well as confirming that upon combining all datasets the individual fields will remain uniquely identifiable. To accomplish this goal the initial challenge was to reformat each individual table from each document in order to have all columns include the category/topic, specific label, and the year, where necessary, so that once each table was combined each column could still be distinguished from one another. Upon completing the initial reformatting of the data and combining them each to form the initial data frame, the next step was to look for gaps within the dataset.

Once identified, we would look to fill the blank/empty cells in the data frame with the median values of the respective column. This led to 144 total cells being filled with their columns median, out of our total 5,300 cells, stating that only 2.7% of our overall model makeup will include data coming from imputation. I chose to add in the median to each blank cell respectively due to this low overall requirement for data to complete the set, as well as the medians ability to avoid possible skews and/or outliers in any specific column.

After imputation was complete, the focus turned to dimensionality reduction, but even more specifically feature selection as tests were run in order to identify and remove all variables that correlated above an 0.8 or eighty percent threshold with another. This is intentionally leaving every variable left within the data frame to have at a minimum, more than twenty percent of its data as unique to itself and lacking correlation to any other variable. This action lowered the total variable count from over 400 to only 106 including our dependent variable.

The final step to prepare our data frame for model testing was to reformat the dependent variable into a dummy/binary variable indicating whether or not the state is ranked within the top 10 overall states. Doing so allows us the opportunity to predict whether or not a state is considered to be in the most desirable tier of states, the top ten.

Outside of the initial data frame, preparation was also done for the model testing in regards to the train-test split ratio. Since the data that is being used within this study is limited to fifty line items since there are only fifty states, it is extremely important to make sure we maximize the testing efficiency and capability of the data, while remaining reliable. In order to accomplish this, I opted to go with a eighty-twenty split in terms of training and test sets, meaning there will be 40 items in our training set, and 10 with the test set.

Data Exploration

Basic Statistics

Raw Counts

Name	Value
Rows	50
Columns	106
Discrete columns	0
Continuous columns	106
All missing columns	0
Missing observations	0
Complete Rows	50
Total observations	5,300
Memory allocation	66.4 Kb

The above table shows the completeness of the final dataset, signifying the fifty states and how each of them are now showing complete in every single data cell within the table. In entirety, this makes up 5,300 total data points that are to be included in model testing.

Additional tables and graphs for this section can be found at the end of the appendix, prior to the data dictionary. There you can find a principal component analysis (figure 1), indicating the best amount of principal components to use within a model, a univariate diagram (figure 2), which assists in identifying trends in each individual variable while also displaying skews and outliers, a QQ plot (figure 5), which displays a visualization of the goodness of fit and allows for variable contrasting, and a correlation analysis graph (figure 4), showing as a heat map for all variables and their similarities/differences with one another.

Summary Statistics

	vars	n	mean	sd	median	trimmed	mad	min	max	range
air_pollution_rank	1	50	24.98	14.56931	23.5	24.9	19.2738	1	50	49
asthma_value	2	50	10.31	1.260061	10.25	10.29	1.11195	7.1	13.9	6.8
behaviors_rank	3	50	25.44	14.57165	25.5	25.425	18.5325	1	50	49
binge_drinking_rank	4	50	24.12	13.80977	24.5	24.075	16.3086	1	48	47
breast_cancer_screening_value	5	50	71.696	4.639975	72.15	71.8225	4.22541	60.9	81.7	20.8
cancer_rank	6	50	23.74	13.67077	23	23.625	16.3086	1	48	47
cannabis_use_rank	7	50	25.36	14.61863	25.5	25.35	18.5325	1	50	49
childhood_immunizations_rank	8	50	25.42	14.50839	25.5	25.425	17.7912	1	50	49
chlamydia_score	9	50	-0.1846	0.963054	-0.15	-0.20925	0.830256	-2	2	4
chronic_kidney_disease_value	10	50	3.82	0.691862	3.75	3.755	0.66717	2.5	6.5	4
climate_policies_score	11	50	-0.0126	1.009483	0.03	-0.022	1.897728	-1.26	1.31	2.57
climate_risks_value	12	50	38.582	25.54812	35.2	37.455	32.6172	1.3	94	92.7
crowded_housing_value	13	50	2.854	1.598878	2.35	2.535	0.66717	1.3	9.4	8.1
dedicated_health_care_provider_ran	14	50	24.4	13.70200	24.5	24.4	17.0499	1	47	46
dental_care_providers_value	15	50	64.094	11.91511	62.25	63.6	10.15581	42.5	97.5	55
dental_visit_value	16	50	65.014	4.324586	65.9	65.075	4.15128	55.6	73.9	18.3
dependency_ages_18_or_64_rank	17	50	24.96	14.41790	25	24.875	18.5325	1	50	49
depression_rank	18	50	24.22	13.57374	24.5	24.25	17.0499	1	48	47
domestic_violence_rank	19	50	25.2	14.57982	25.5	25.2	19.2738	1	50	49

Above is a truncated table of the summary statistics for the variables included in the data frame utilized for model testing. Here you are able to see some key indicators for these fields and their own unique trends, skews, ranges and overall statistical behavior. For many of the rank variables you will see a range of 49 as the states are ranked one through fifty, but in some cases if there were ties within rankings, or imputation was required as a state was not ranked due to lack of credible information, the range may appear to be smaller than the expected value.

Models

Algorithm(s) Selection

Multiple different model types were tested as a part of the research and side by side testing in order to identify the best performing models to proceed with. The tested model types included the below and their respective initial testing results.

Model	Accuracy	P-Value [Acc > NIR]	Balanced Accuracy
LDA	0.9	0.3758	0.75
GLM	0.7	0.8791	0.8125
RandomForrest	0.8	0.6778	0.5
Multinom	1	0.1074	1
Neuralnet	1	0.1074	1

As you can see from the above graph, the results indicated that there were clearly two models that were our best performing, yet this was only from the initial model results and had so far only included k-fold cross validation in terms of adaptations from the initial models setup. To ensure that each of these well performing models was not memorizing the noise in the training set nor overfitting I implemented the following changes to each of them, in order to fine tune and verify which model truly performs the best.

Multinomial Regression Adaptions:

1. **Adding Weight Decay.** By adding weight decay we are reducing potential overfitting and preventing the model from memorizing the noise within the training data.
2. **Adding the Select Function**. Another step taken in order to reduce overfitting, adding the selectFunction = “oneSE” in our train_control step helps by opting for the most simplified model that performed closest to, or is, the best performing model.
3. **Adding Near Zero Variance**. By breaking out the preprocess detail of the model and adding in the specification regarding near zero variance, we look to continue reducing possible overfitting and increasing the overall stability of the model.
4. **Increasing Iterations**. Within the model I increased the iterations in order to ensure the model does converge.
5. **Adapting the Summary Function**. After some due diligence it appeared that the multiClassSummary option is much more reliable and robust, making it the preferred summary function for our model testing.

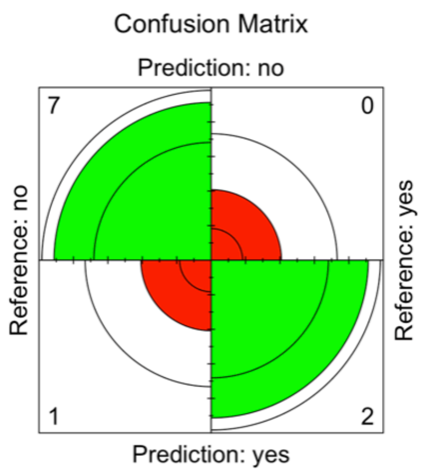
Neural Network Regression Adaptations:

1. **Adjusting the Weight Decay and Sizing.** By adding weight decay we are reducing potential overfitting and preventing the model from memorizing the noise within the training data. I also limited sizes across the model, assisting the model in creating more generalization across all of the independent variables.
2. **Adding the Select Function**. Another step taken in order to reduce overfitting, adding the selectFunction = “oneSE” in our train_control step helps by opting for the most simplified model that performed closest to, or is, the best performing model.

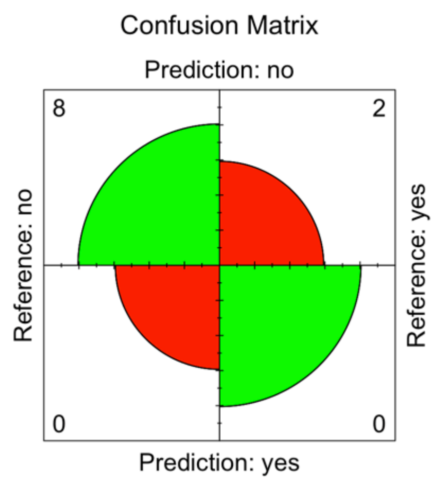
3. **Adding Near Zero Variance**. By breaking out the preprocess detail of the model and adding in the specification regarding near zero variance, we look to continue reducing possible overfitting and increasing the overall stability of the model. Neural networks are especially grateful for this in their processing as it eliminates invalid significance being assigned to noise.
4. **Increasing Iterations**. Within the model I increased the iterations in order to ensure the model does complete the entirety of the training process.
5. **Adapting the Summary Function**. After some due diligence it appeared that the multiClassSummary option is much more reliable and robust, making it the preferred summary function for our model testing.

The implemented changes to each individual model assisted in controlling the model behavior and downstream results, leading to more reliable processing by each model. This provides us with a much more accurate representation of the models true ability to predict our dependent variable, and tell us whether or not a state is considered to be top 10. Results from the model tests after fine tuning each model is shown below.

Multinomial Fine-Tuned Model:



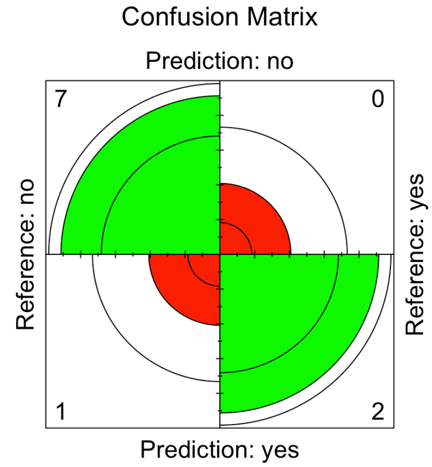
Neural Network Fine-Tuned Model:



As you can see from the above results, the multinomial model now clearly performs better than the neural network model once we have implement the many controls and in turn drastically reduced potential overfitting within each model.

	Training Accuracy	Testing Accuracy	Gap
Neural Network Model	0.8	0.8	0
Multinomial Model	1	0.9	0.1

Final Model



As displayed in the end of the previous section the best performing model in being able to predict whether or not a state would be considered to be top ten was the Multinomial model. The results for the model showed an accuracy of 90%, with a 95% confidence interval of 55.5% to 99.75%, a balanced accuracy of 93.75%, and a sensitivity of 87.5%. The visual for the performance of the model can be found to the right.

In interpreting the model results we will go through each resulting significant statistic. The accuracy of the model, which was 90%, at first glance would appear to be amazing, but when taking into the account the no information rate is 80%, we are barely beating the minimum threshold of just guessing no every time. Many variables can come into play regarding these resulting statistics, but in this scenario it is important to make note of sample size and that if we were to run again, there could be a high likelihood one of the predictions flips, leading to a 10% or more shift in our reported accuracy, removing the incentive of using the model at all. The P-value [Acc > NIR] of 37.58% indicates similarly that our model is not better than the initial 80% baseline that would be met if you just assumed all states were not top ten.

The reported Kappa value of 73.68% indicates to us that the model itself did identify and learn reliable patterns within the training portion of the model setup, as it means that the model itself performed with a 73.68% reduction in overall errors in contrast to that of a chance based prediction model.

The specificity, sensitivity, positive and negative predication values all are statistical representations of the models ability to predict each class, yes or no is a state considered top 10, correctly. What the data results tell us is that this model is reliable in predicting no, but also that it is not reliable enough in predicting overall whether or not a state is considered top ten due to the high P-value present in the produced confusion matrix.

Due to our results from the model, we will reject our null hypothesis, and state rather that we are currently unable to confidently predict whether or not a state would be considered top 10 using the fields included within this study.

In order for the model to prove reliable and statistically significant, far more data will be required to add to testing and training the model. Since the constructed dataset is limited by how many states there are, the future approach in expanding the dataset may require some backtracking and reformatting how the dataset is constructed and combined. As future years go on, and more data is collected and states are reranked year over year, instead of overwriting the line items for each state from the previous years with the new data the new years data can be added using the same fields and sources but as new line items. After one year that will allow for the dataset to go from 50 line items to 100 total with then having 20 of them being considered top ten states. Each year this dataset will increase again by fifty, which then means each year the dataset will increase in datapoints, helping our training and testing process by providing more information to learn and trends to analyze.

Discussion & Next Steps

To summarize the research study completed and described in this exploration, we were unable to confidently predict whether or not a state is considered to be top ten according to the U.S. News article. What is important to take note of here is our dependent variable, though constructed from a combination of government sources and trusted data providers, is not a concrete ranking. It is a subject collaboration of what the U.S. new determined to be indicators of a successful state. The ranking itself is extremely subjective, and it is rather likely that very few people would have the same exact rankings of the 50 states from their perspective. Irregardless the general rankings they provided were unable to be predicted reliably. In an alternate study where we could accurately predict the general rankings, for it to be a reliable model for the public you would likely also want to include binary indicators, filled out by the person utilizing the model that would customize the results. For example, whether or not they are from a specific state and/or country, whether or not they speak multiple languages/which languages they speak, and even more binary questions for them to indicate their unique preferences to adapt the resulting model.

In terms of our model itself and its performance, although we were forced to reject the null hypothesis in favor of the alternative, it does not mean that a lesson was not learned and that this information could not be taken to then improve upon this data setup and modeling procedure to continue to search for significant model results. Our kappa value did indicate that the model training portion was successful in picking up trends within the dataset, assisting the model in being able to locate top 10 states. As mentioned in the previous section, the core focus moving forward is to minimize the P-value, by providing additional year over year data for these states as additional line items / as if they were new states. Doing so will increase the overall size of the dataset greatly, providing even more data points to train on, test with, and identify trends within.

Code Availability Section - Github Link

https://github.com/ajohnstone34/Desirable-US-Living-States/tree/main

Appendix

Figure 1

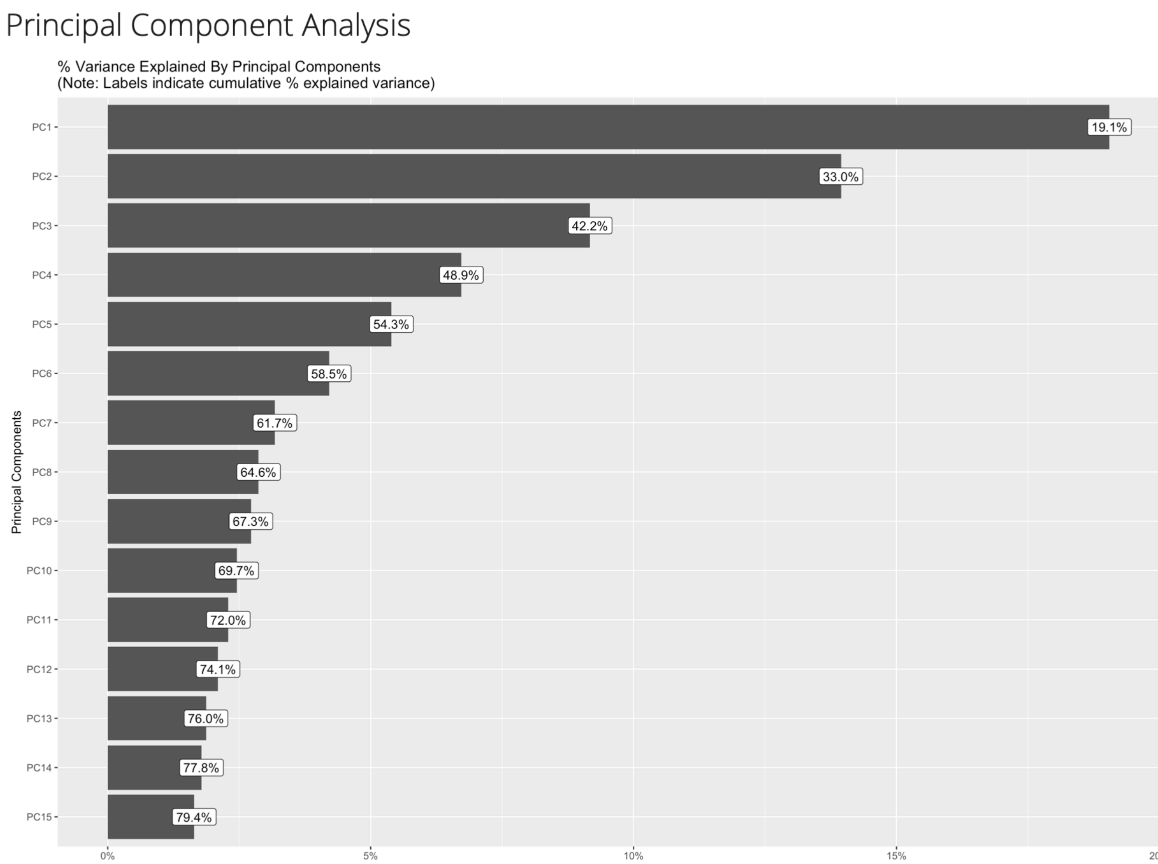


Figure 2

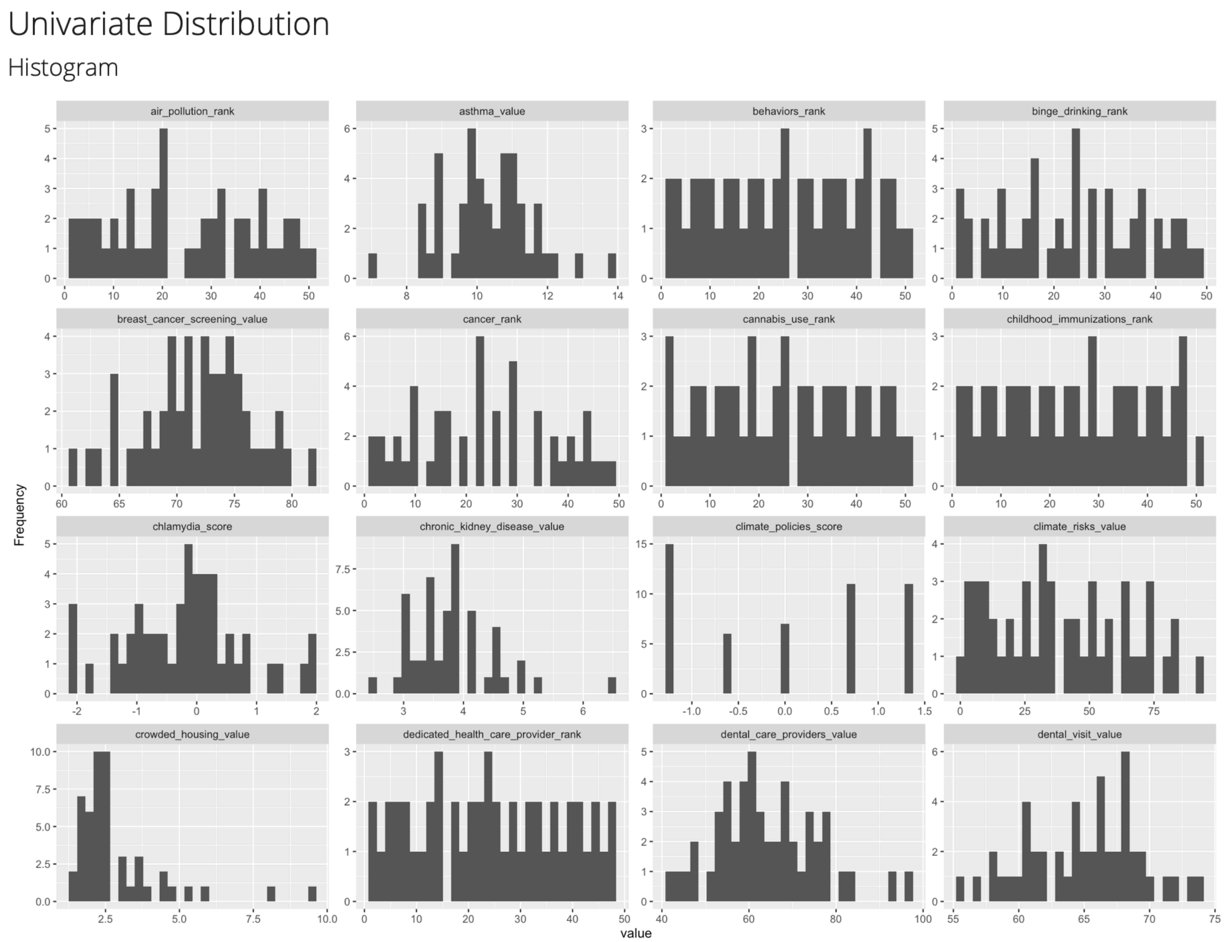


Figure 3

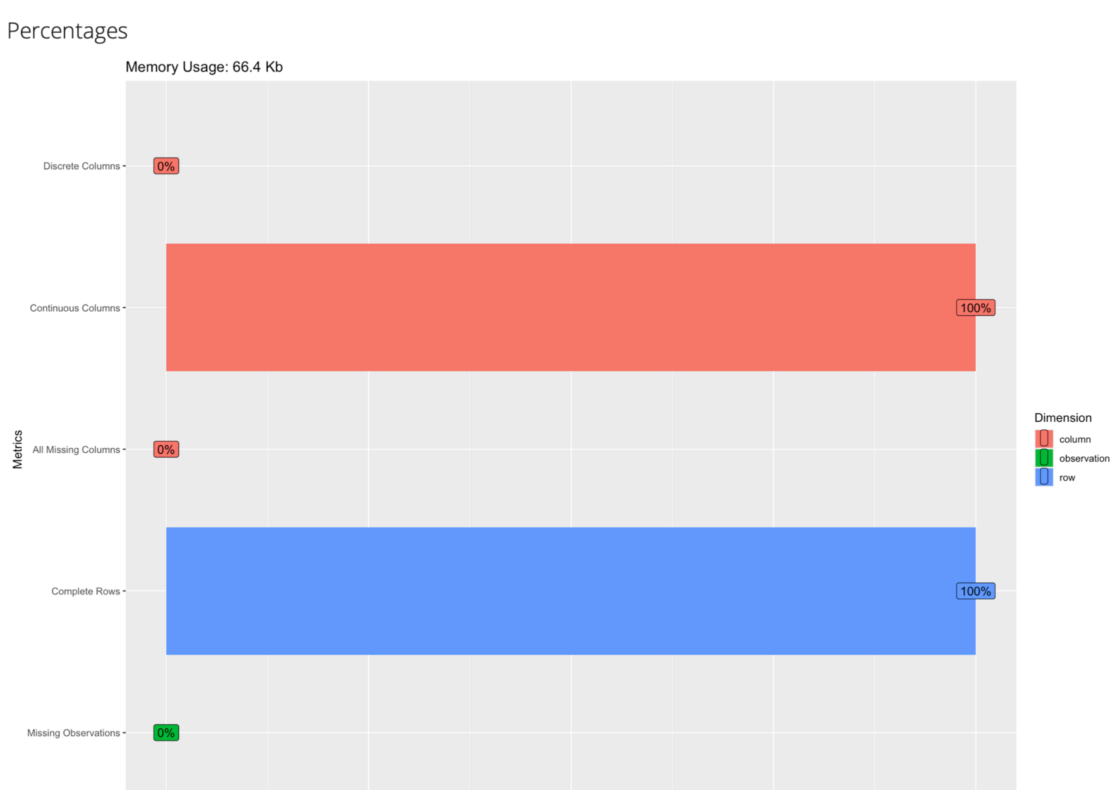


Figure 4

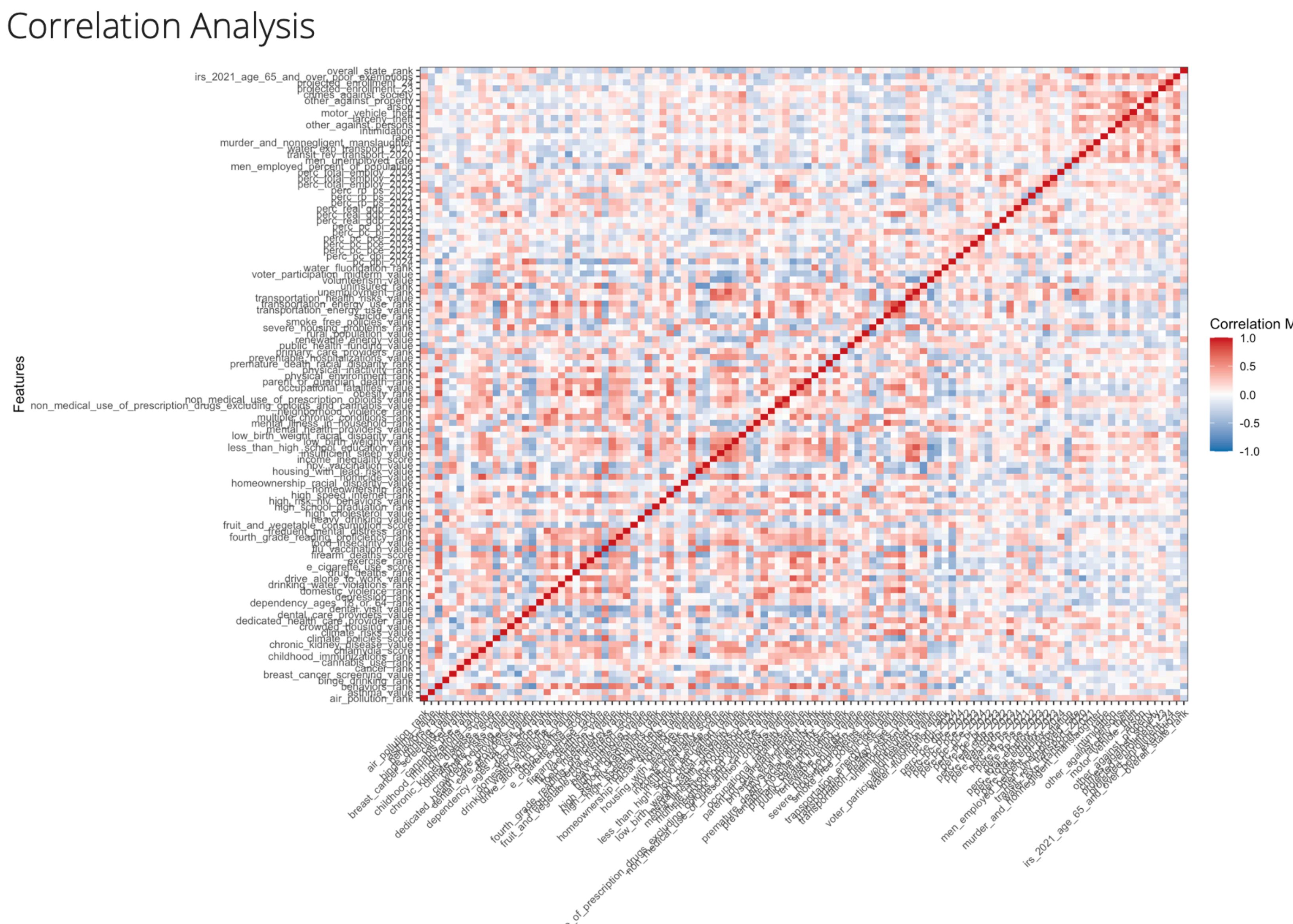
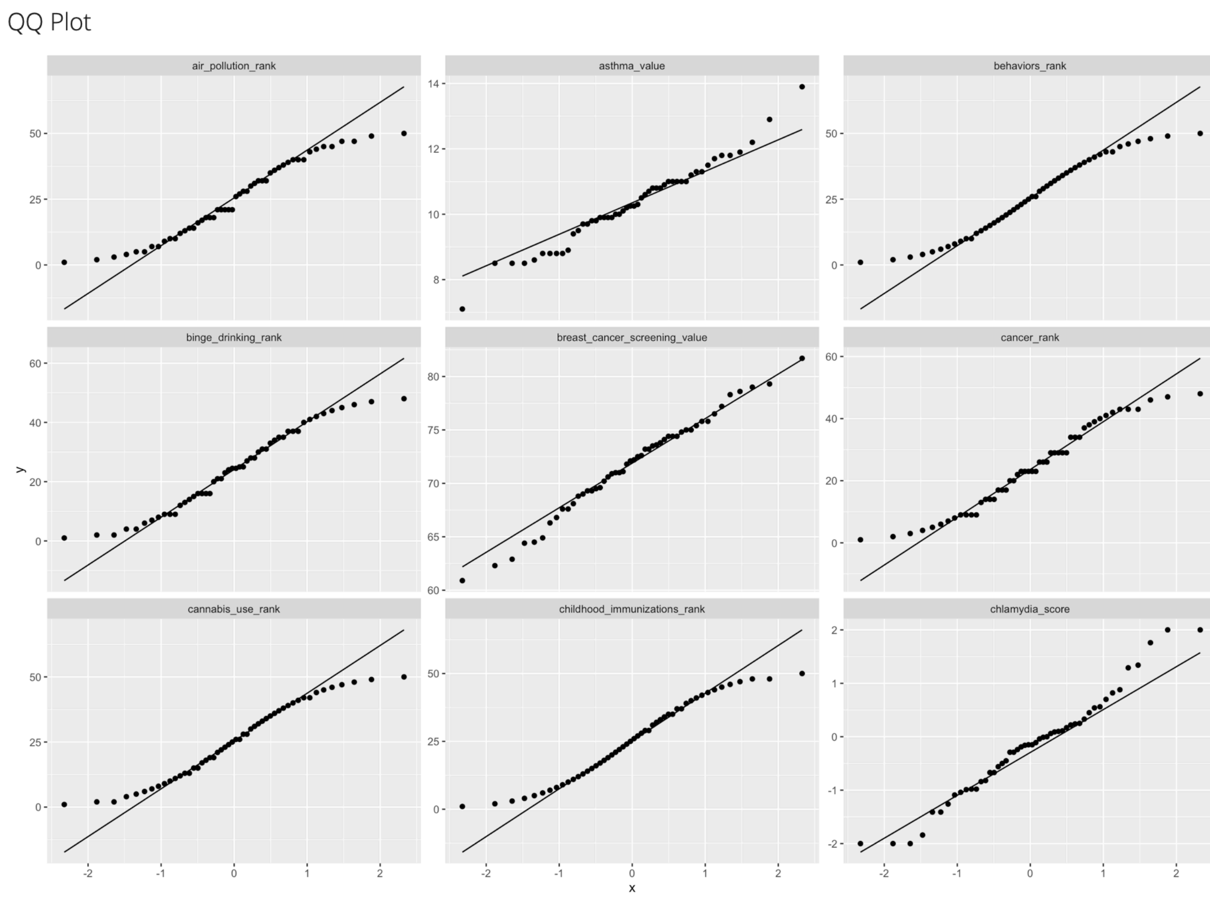

Correlation Analysis
Features
Correlation M
1.0
0.5
0.0
-0.5
-1.0
overall_state_rank
irs_2021_age_65_and_over_poor_exemptions
crimes_against_society
other_against_property
arson
motor_vehicle_theft
other_against_persons
murder_and_nonnegligent_manslaughter
men_unemployed_rate
men_employed_percent_of_population
water_fluoridation_rank
voter_participation_midterm_value
volunteerism_value
uninsured_rank
unemployment_rank
transportation_health_risks_value
transportation_energy_use_rank
transportation_energy_use_value
smoke_free_policies_value
severe_housing_problems_rank
rural_population_value
renewable_energy_value
public_health_funding_value
primary_care_providers_rank
preventable_hospitalizations_value
premature_death_racial_disparity_rank
physical_environment_rank
parent_to_guardian_health_rank
occupational_fatalities_value
non_medical_use_of_prescription_opioids_rank
non_medical_use_of_prescription_drugs_excluding_opioids_and_cannabis_value
neighborhood_violence_rank
multiple_chronic_conditions_rank
mental_health_providers_value
low_birth_weight_racial_disparity_rank
low_birth_weight_value
less_than_high_school_education_rank
insufficient_sleep_value
income_inequality_score
hpv_vaccination_value
housing_with_lead_risk_value
homeownership_racial_disparity_value
homeownership_rank
high_speed_internet_rank
high_risk_hiv_behaviors_value
high_school_graduation_rank
high_cholesterol_value
heavy_drinking_score
fruit_and_vegetable_consumption_score
frequent_mental_distress_rank
fourth_grade_reading_proficiency_rank
food_insecurity_value
flu_vaccination_value
firearm_deaths_score
exercise_rank
e_cigarette_use_score
drug_deaths_rank
drive_alone_to_work_value
drinking_water_violations_rank
domestic_violence_rank
depression_rank
dependency_ages_18_to_64_value
dental_visit_value
dental_care_providers_value
dedicated_health_care_provider_rank
crowded_housing_value
climate_risks_value
chronic_kidney_disease_value
chlamydia_score
childhood_immunizations_rank
cannabis_use_rank
cancer_rank
breast_cancer_screening_value
binge_drinking_rank
behaviors_rank
asthma_value
air_pollution_rank

Figure 5


QQ Plot
air_pollution_rank
asthma_value
behaviors_rank
binge_drinking_rank
breast_cancer_screening_value
cancer_rank
cannabis_use_rank
childhood_immunizations_rank
chlamydia_score
y
x

Data Dictionary

Field Name	Source(s)	Definition
air_pollution_rank	2021 - 2023 U.S. Environmental Protection Agency	State ranking by air pollution rate
asthma_value	2023 CDC, Behavioral Risk Factor Surveillance System	Numerical representation of a state’s asthma rate
behaviors_rank	2024 America's Health Rankings composite measure	State ranking by behavior quality
binge_drinking_rank	2023 CDC, Behavioral Risk Factor Surveillance System	State ranking by binge drinking rate
breast_cancer_screening_value	2022 CDC, Behavioral Risk Factor Surveillance System	Numerical representation of a state's breast cancer screening rate
cancer_rank	2023 CDC, Behavioral Risk Factor Surveillance System	State ranking by cancer rate
cannabis_use_rank	2024 Denver Health and Hospital Authority, RADARS System Survey of Non-Medical Use of Prescription Drugs Program	State ranking by cannabis use rate
childhood_immunizations_rank	2020 - 2021 Birth Cohort, CDC, National Immunization Survey-Child	State ranking by childhood immunization rate
chlamydia_score	2022 CDC, National Center for HIV/AIDS, Viral Hepatitis, STD, and TB Prevention Atlas	Numerical representation of a state's chlamydia rate
chronic_kidney_disease_value	2023 CDC, Behavioral Risk Factor Surveillance System	Numerical representation of a state's chronic kidney disease rate
climate_policies_score	2023 Center for Climate and Energy Solutions	Numerical representation of the quality of a state's climate policies
climate_risks_value	2022 Council on Environmental Quality, Climate and Economic Justice Screening Tool Index	Numerical representation of a state's climate risks
crowded_housing_value	2023 U.S. Census Bureau, American Community Survey	Numerical representation of a state's crowded housing rate
dedicated_health_care_provider_rank	2023 CDC, Behavioral Risk Factor Surveillance System	State ranking by dedicated health care provider quality

Field Name	Source(s)	Definition
dental_care_providers_value	2024 U.S. HHS, Centers for Medicare & Medicaid Services, National Plan and Provider Enumeration System	Numerical representation of a state’s dental care providers quality
dental_visit_value	2022 CDC, Behavioral Risk Factor Surveillance System	Numerical representation of a state's dental visit rate
dependency_ages_18_or_64_rank	2023 U.S. Census Bureau, American Community Survey	State ranking by dependency for people who are between the ages of 18 and 64
depression_rank	2023 CDC, Behavioral Risk Factor Surveillance System	State ranking by depression rate
domestic_violence_rank	2022 - 2023 National Survey of Children's Health, U.S. Department of Health and Human Services, Health Resources and Services Administration (HRSA), Maternal and Child Health Bureau (MCHB)	State ranking by domestic violence rate
drinking_water_violations_rank	2023 U.S. Environmental Protection Agency, Enforcement and Compliance History Online, Safe Drinking Water Information System	State ranking by drinking water violations
drive_alone_to_work_value	2023 U.S. Census Bureau, American Community Survey	Numerical representation of a state's drive alone to work rate
drug_deaths_rank	2022 CDC WONDER, Multiple Cause of Death Files	State ranking by drug deaths rate
e_cigarette_use_score	2023 CDC, Behavioral Risk Factor Surveillance System	Numerical representation of a state's e-cigarette use
exercise_rank	2023 CDC, Behavioral Risk Factor Surveillance System	State ranking by exercise rate
firearm_deaths_score	2022 CDC WONDER, Multiple Cause of Death Files	Numerical representation of a states firearm deaths
flu_vaccination_value	2023 CDC, Behavioral Risk Factor Surveillance System	Numerical representation of a state's flu vaccination rate
food_insecurity_value	2021 - 2023 U.S. Department of Agriculture, Household Food Security in the United States Report Series	Numerical representation of a state's food insecurity rate

Field Name	Source(s)	Definition
fourth_grade_reading_proficiency_rank	2022 U.S. Department of Education, National Center for Education Statistics, National Assessment of Educational Progress	State ranking by fourth grade reading proficiency
frequent_mental_distress_rank	2023 CDC, Behavioral Risk Factor Surveillance System	State ranking by frequent mental distress rate
fruit_and_vegetable_consumption_score	2021 CDC, Behavioral Risk Factor Surveillance System	Numerical representation of a state's fruit and vegetable consumption rate
heavy_drinking_value	2023 CDC, Behavioral Risk Factor Surveillance System	Numerical representation of a state's heavy drinking rate
high_cholesterol_value	2023 CDC, Behavioral Risk Factor Surveillance System	Numerical representation of a states population's high cholesterol rate
high_school_graduation_rank	2021 - 2022 U.S. Department of Education, National Center for Education Statistics, Common Core of Data	State ranking by high school graduation rate
high_risk_hiv_behaviors_value	2022 CDC, Behavioral Risk Factor Surveillance System	Numerical representation of a state's high risk HIV behaviors
high_speed_internet_rank	2023 U.S. Census Bureau, American Community Survey	State ranking by high speed internet
homeownership_rank	2023 U.S. Census Bureau, American Community Survey	State ranking by homeownership rate
homeownership_racial_disparity_value	2023 U.S. Census Bureau, American Community Survey	Numerical representation by racial disparity of homeownership rate
homicide_value	2021 - 2022 CDC WONDER, Multiple Cause of Death Files	Numerical representation of a state's homicide rate
housing_with_lead_risk_value	2023 U.S. Census Bureau, American Community Survey	Numerical representation of a state's housing with lead risk
hpv_vaccination_value	2023 CDC, National Immunization Survey-Teen	Numerical representation of a state's HPV vaccination rate
income_inequality_score	2023 U.S. Census Bureau, American Community Survey	Numerical representation of a state's income inequality
insufficient_sleep_value	2022 CDC, Behavioral Risk Factor Surveillance System	Numerical representation of a state's average inefficient sleep
less_than_high_school_education_rank	2023 U.S. Census Bureau, American Community Survey	State ranking by people who have achieved less than a full high school education

Field Name	Source(s)	Definition
low_birth_weight_value	2022 CDC WONDER, Natality Public Use Files	Numerical representation of a state's low birth weight
low_birth_weight_racial_disparity_rank	2020 - 2022 CDC WONDER, Natality Public Use Files	State ranking by racial disparity of low birth weights
mental_health_providers_value	2024 U.S. HHS, Centers for Medicare & Medicaid Services, National Plan and Provider Enumeration System	Numerical representation of a state's quality of their mental health providers
mental_illness_in_household_rank	2022 - 2023 National Survey of Children's Health, U.S. Department of Health and Human Services, Health Resources and Services Administration (HRSA), Maternal and Child Health Bureau (MCHB)	State ranking by mental illness within households
multiple_chronic_conditions_rank	2023 CDC, Behavioral Risk Factor Surveillance System	State ranking by prevalence of multiple chronic conditions
neighborhood_violence_rank	2022 - 2023 National Survey of Children's Health, U.S. Department of Health and Human Services, Health Resources and Services Administration (HRSA), Maternal and Child Health Bureau (MCHB)	State ranking by neighborhood violence
non_medical_use_of_prescription_drugs_excluding_opioids_and_cannabis_value	2024 Denver Health and Hospital Authority, RADARS System Survey of Non-Medical Use of Prescription Drugs Program	Numerical representation of a state's non-medical use of prescription excluding opioids and cannabis
non_medical_use_of_prescription_opioids_value	2024 Denver Health and Hospital Authority, RADARS System Survey of Non-Medical Use of Prescription Drugs Program	Numerical representation of a state's non-medical use of prescription opioids
obesity_rank	2023 CDC, Behavioral Risk Factor Surveillance System	State ranking by obesity
occupational_fatalities_value	2020 - 2022 U.S. Bureau of Labor Statistics, Census of Fatal Occupational Injuries	Numerical representation of a state’s occupational fatalities

Field Name	Source(s)	Definition
parent_or_guardian_death_rank	2022 - 2023 National Survey of Children's Health, U.S. Department of Health and Human Services, Health Resources and Services Administration (HRSA), Maternal and Child Health Bureau (MCHB)	State ranking by deaths of parents and/pr guardians
physical_environment_rank	2024 America's Health Rankings composite measure	State ranking by physical environment
physical_inactivity_rank	2023 CDC, Behavioral Risk Factor Surveillance System	State ranking by physical inactivity
premature_death_racial_disparity_rank	2018 - 2020 CDC WONDER, Multiple Cause of Death Files	State ranking by the racial disparity of premature deaths
preventable_hospitalizations_value	2022 U.S. HHS, Centers for Medicare & Medicaid Services, Office of Minority Health, Mapping Medicare Disparities Tool	Numerical representation of a state's preventable hospitalizations
primary_care_providers_rank	2024 U.S. HHS, Centers for Medicare & Medicaid Services, National Plan and Provider Enumeration System	State ranking by the quality of primary care providers
public_health_funding_value	2022 - 2023 CDC, HRSA and Trust for America's Health	Numerical representation of a state's public health funding
renewable_energy_value	2023 U.S. Energy Information Administration, State Energy Data System	Numerical representation of a state's renewable energy sector
rural_population_value	2023 U.S. Census Bureau, American Community Survey	Numerical representation of a state's rural population
severe_housing_problems_rank	2017 - 2021 U.S. Department of Housing and Urban Development, Comprehensive Housing Affordability Strategy	State ranking of prevalence of severe housing problems
smoke_free_policies_value	2024 American Nonsmokers' Rights Foundation	Numerical representation of state policies for smoke free areas
suicide_rank	2022 CDC WONDER, Multiple Cause of Death Files	State ranking by suicide rate
transportation_energy_use_value	2022 U.S. Energy Information Administration, State Energy Data System	Numerical representation of the energy usage for the purpose of transportation

Field Name	Source(s)	Definition
transportation_energy_use_rank	2022 U.S. Energy Information Administration, State Energy Data System	State ranking of energy use for the purpose of transportation
transportation_health_risks_value	2022 Council on Environmental Quality, Climate and Economic Justice Screening Tool Index	Numerical representation of the health risks using transportation in a specific state
unemployment_rank	2023 U.S. Census Bureau, American Community Survey	State ranking by unemployment rate
uninsured_rank	2023 U.S. Census Bureau, American Community Survey	State ranking by percent of uninsured people
volunteerism_value	2021 U.S. Census Bureau, Current Population Survey, Volunteering and Civic Life Supplement	Numerical representation of volunteer percent at the state level
voter_participation_midterm_value	2020 - 2022 U.S. Census Bureau, Current Population Survey, Voting and Registration Supplement	Numerical representation of the state voter participation in the midterm
water_fluoridation_rank	2022 CDC, Water Fluoridation Reporting System	State ranking by water fluoridation quality
pc_dpi_2024	U.S. Bureau of Economic Analysis, Regional Data Interactive Data Tables	Per capita disposable personal income
perc_pc_dpi_2024	U.S. Bureau of Economic Analysis, Regional Data Interactive Data Tables	Percent change in per capita disposable personal income
perc_pc_pce_2022	U.S. Bureau of Economic Analysis, Regional Data Interactive Data Tables	Percent change in per capita personal consumption expenditures
perc_pc_pce_2023	U.S. Bureau of Economic Analysis, Regional Data Interactive Data Tables	Percent change in per capita personal consumption expenditures
perc_pc_pce_2024	U.S. Bureau of Economic Analysis, Regional Data Interactive Data Tables	Percent change in per capita personal consumption expenditures
perc_pc_pi_2022	U.S. Bureau of Economic Analysis, Regional Data Interactive Data Tables	Percent change in per capita personal income
perc_pc_pi_2023	U.S. Bureau of Economic Analysis, Regional Data Interactive Data Tables	Percent change in per capita personal income

Field Name	Source(s)	Definition
perc_real_gdp_2022	U.S. Bureau of Economic Analysis, Regional Data Interactive Data Tables	Percent change in real GDP
perc_real_gdp_2023	U.S. Bureau of Economic Analysis, Regional Data Interactive Data Tables	Percent change in real GDP
perc_real_gdp_2024	U.S. Bureau of Economic Analysis, Regional Data Interactive Data Tables	Percent change in real GDP
perc_rp_ps_2021	U.S. Bureau of Economic Analysis, Regional Data Interactive Data Tables	Percent change in regional price parities in 2021
perc_rp_ps_2022	U.S. Bureau of Economic Analysis, Regional Data Interactive Data Tables	Percent change in regional price parities in 2022
perc_rp_ps_2023	U.S. Bureau of Economic Analysis, Regional Data Interactive Data Tables	Percent change in regional price parities in 2023
perc_total_employ_2022	U.S. Bureau of Economic Analysis, Regional Data Interactive Data Tables	Percent change in total employment in 2022
perc_total_employ_2023	U.S. Bureau of Economic Analysis, Regional Data Interactive Data Tables	Percent change in total employment in 2023
perc_total_employ_2024	U.S. Bureau of Economic Analysis, Regional Data Interactive Data Tables	Percent change in total employment in 2024
men_employed_percent_of_population	2024 U.S. Bureau of Labor Statistics, Expanded State Employment Status Demographic Data	Percent of the total male population that is employed
men_unemployed_rate	2024 U.S. Bureau of Labor Statistics, Expanded State Employment Status Demographic Data	Percent of the total male population that is unemployed
transit_rev_transport_2020	Bureau of Transportation Statistics, Transportation Revenues and Expenditures Table updated to 2024	Transit state and local revenue as of 2020
water_exp_transport_2021	Bureau of Transportation Statistics, Transportation Revenues and Expenditures Table updated to 2024	Water state and local expenditures as of 2021

Field Name	Source(s)	Definition
murder_and_nonnegligent_manslaughter	2024 FBI Crime Data Explorer, Uniform Crime Report	Count of murder and non negligent manslaughter crimes
rape	2024 FBI Crime Data Explorer, Uniform Crime Report	Count of rape crimes
intimidation	2024 FBI Crime Data Explorer, Uniform Crime Report	Count of intimidation crimes
other_against_persons	2024 FBI Crime Data Explorer, Uniform Crime Report	Count of crimes against people that did not fit a specific category
larceny_theft	2024 FBI Crime Data Explorer, Uniform Crime Report	Count of larceny crimes
motor_vehicle_theft	2024 FBI Crime Data Explorer, Uniform Crime Report	Count of motor vehicle theft crimes
arson	2024 FBI Crime Data Explorer, Uniform Crime Report	Count of arson crimes
other_against_property	2024 FBI Crime Data Explorer, Uniform Crime Report	Count of crimes against property that did not fit a specific category
crimes_against_society	2024 FBI Crime Data Explorer, Uniform Crime Report	Count of crimes against society
projected_enrollment_23	National Center for Education Statistics, Table 203.20	Projected enrollment in public elementary and secondary schools
projected_enrollment_24	National Center for Education Statistics, Table 203.20	Projected enrollment in public elementary and secondary schools
irs_2021_age_65_and_over_poor_exemptions	2024 United States Census Bureau SAIPE Model Input Data	Count of IRS exemptions for people age 65 and over in 2021
overall_state_rank	2025 U.S News Best States Rankings	Overall ranking of states

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